Domain-Adaptive Dense Retrieval for Short Scientific Questions: A Controlled Benchmark

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Abstract

This fictional study tests domain-adaptive training for dense retrieval over the same 1,200-pair synthetic research-question benchmark used in a companion demonstration paper. The evaluation uses the same 960-pair development set, 240-pair test set, passage collection, and relevance labels. A dual encoder adapted on development pairs reaches NDCG@10 of 0.768 and Recall@10 of 0.826, compared with 0.681 and 0.754 for a BM25 baseline. Median latency is 240 ms per query. The point estimates exceed the baseline on both ranking metrics, while latency is higher. Because training and evaluation are synthetic, results use only one test split, and there is no external replication, this study supports only a within-benchmark comparison and not a general conclusion about domain adaptation.

1. Introduction

Dense retrieval can match semantically related passages even when query and document wording differs. Its performance may depend on how closely training examples represent the target domain. This demonstration evaluates a dual encoder adapted on labeled development questions and compares it with a lexical baseline on a fixed synthetic test set.

Research question: How do the tested retrieval methods compare on ranking quality and latency in this fixed synthetic benchmark?

2. Methods

The baseline is BM25 with k1 = 1.2 and b = 0.75. The adapted system uses a dual encoder initialized from a general text model and fine-tuned on positive and sampled negative question-passage pairs from the 960-pair development set. It returns the top 10 passages by cosine similarity. The 240-pair test set is held out from adaptation. No test examples are used for training or parameter selection.

3. Dataset and evaluation

The benchmark contains the same 1,200 synthetic question-document pairs across computing, biology, materials science, and public health. The development set contains 960 pairs and the held-out test set contains 240 pairs. Each question has between one and five relevant passages. Values are illustrative and were authored for software demonstration; the benchmark is not a published or externally validated dataset.

We report NDCG@10 and Recall@10 for ranking quality, plus median per-query latency. Both ranking metrics range from 0 to 1, with higher values indicating better performance. Latency is measured in milliseconds, with lower values indicating faster responses. No uncertainty interval is available.

4. Results

System	NDCG@10	Recall@10	Median latency
BM25 baseline	0.681	0.754	92 ms
Domain-adapted dual encoder	0.768	0.826	240 ms

The adapted dual encoder improves NDCG@10 by 0.087 and Recall@10 by 0.072 relative to BM25 in this test split. Its median latency is 240 ms, compared with 92 ms for the baseline. These are descriptive point estimates. The study does not report confidence intervals or statistical significance tests, so the observed differences may not be stable across other samples.

5. Limitations

The data and relevance labels are synthetic, and adaptation uses examples drawn from the same constructed benchmark. Only one fixed test split and one encoder configuration are evaluated. We did not test external document collections, real researcher behavior, multilingual data, robustness to domain shift, or answer-level quality. Direct comparison with other methods requires a shared implementation and compute setup.

6. Conclusion

On this fictional test split, domain adaptation has higher ranking scores and higher latency than BM25. Further evaluation on independently collected scholarly queries, repeated splits, uncertainty estimates, and measured deployment hardware would be needed before making a practical recommendation.

Data and provenance note

All study names, authors, data, metrics, and findings in this PDF are fictional and generated solely for application testing. They must not be cited as real research.